lab2.1.go

```go
package main

import "fmt"

func main() {
    fmt.Println("=== Slice operations ===")
    students := []string{"Asha", "Bhavya", "Chirag"}
    fmt.Println("Initial:", students)

    students = append(students, "Divya")
    fmt.Println("After add 'Divya':", students)

    index := 1
    students = append(students[:index], students[index+1:]...)
    fmt.Println("After remove index 1:", students)

    students[2] = "Esha"
    fmt.Println("After update index 2 to 'Esha':", students)
    fmt.Println("Length:", len(students), "Capacity:", cap(students))

    fmt.Println("\n=== Map operations ===")
    marks := map[string]int{"Maths": 85, "Physics": 78}
    fmt.Println("Initial:", marks)

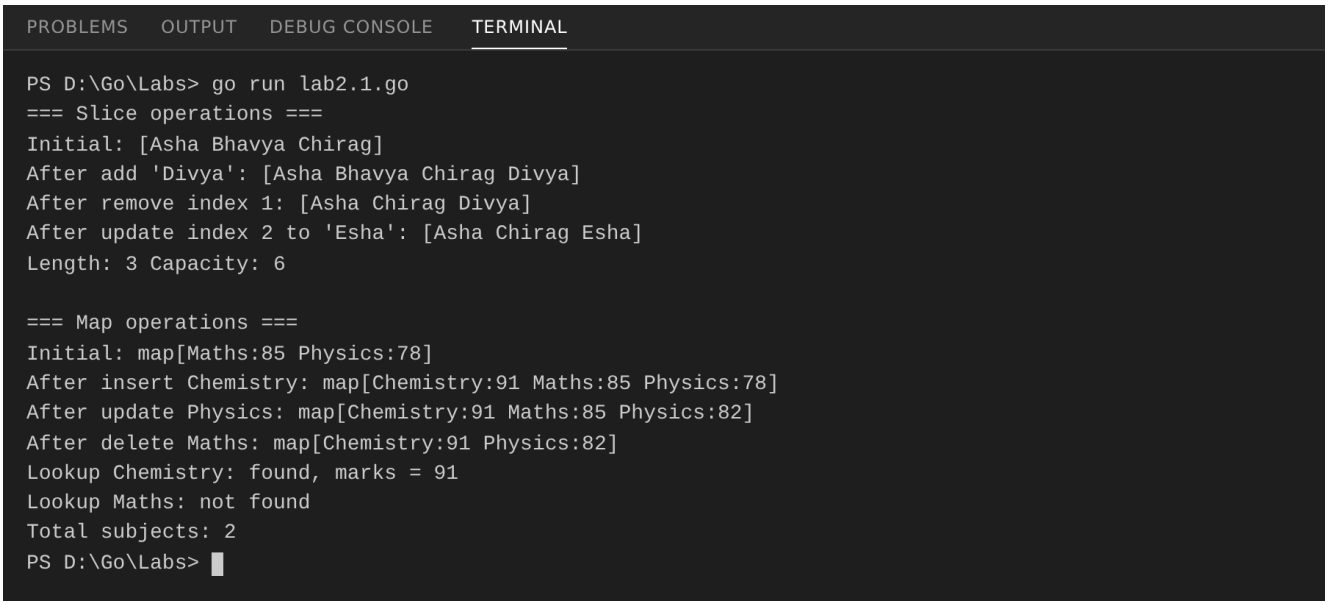
    marks["Chemistry"] = 91
    fmt.Println("After insert Chemistry:", marks)

    marks["Physics"] = 82
    fmt.Println("After update Physics:", marks)

    delete(marks, "Maths")
    fmt.Println("After delete Maths:", marks)

    subjects := []string{"Chemistry", "Maths"}
    for _, s := range subjects {
        if value, ok := marks[s]; ok {
            fmt.Printf("Lookup %s: found, marks = %d\n", s, value)
        } else {
            fmt.Printf("Lookup %s: not found\n", s)
        }
    }
    fmt.Println("Total subjects:", len(marks))
}
```

Output


PROBLEMS
OUTPUT
DEBUG CONSOLE
TERMINAL
PS D:\Go\Labs> go run lab2.1.go
=== Slice operations ===
Initial: [Asha Bhavya Chirag]
After add 'Divya': [Asha Bhavya Chirag Divya]
After remove index 1: [Asha Chirag Divya]
After update index 2 to 'Esha': [Asha Chirag Esha]
Length: 3 Capacity: 6
=== Map operations ===
Initial: map[Maths:85 Physics:78]
After insert Chemistry: map[Chemistry:91 Maths:85 Physics:78]
After update Physics: map[Chemistry:91 Maths:85 Physics:82]
After delete Maths: map[Chemistry:91 Physics:82]
Lookup Chemistry: found, marks = 91
Lookup Maths: not found
Total subjects: 2
PS D:\Go\Labs>